

YEAR 7 MATHEMATICS

Chapter 11

Probability

MIXED PRACTICE WORKSHEETS

11

How to use this booklet

- Each lesson is **one double-sided sheet**. Write your name, class and date at the top.
- Then work down the **columns**, left to right. They get harder as you go: **Easy** **Easy** **Medium** on page 1, then **Medium** **Challenging** **Challenging** on page 2.
- Most questions practise **today's skill**. A question marked **from 1.03** comes from an earlier lesson, to keep it fresh. Stuck? Look back at that lesson.
- Each question tells you what to do: read it carefully. Show your working on the lines. The first question with working in each Easy column shows you how to set it out.

WE DO

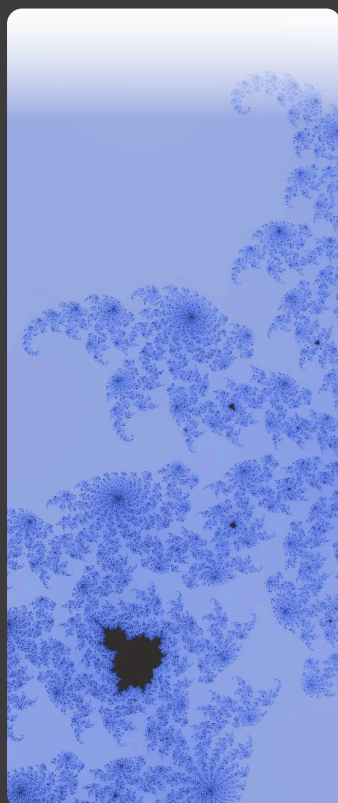
The grey box at the top. Your teacher shows the key idea with two or three examples. **Copy every step** into the space.

Contents and progress tracker

Lesson	Topic	Page	Date	Done
11.01	Sample space	3		☐
11.02	Probability	5		☐
11.03	The probability scale	7		☐
11.04	Experimental probability	9		☐
11.05	Complementary events	11		☐

In this chapter you will

- list the sample space of an experiment
- find the probability of simple events as fractions, decimals and percentages
- use the probability scale and chance words
- compare experimental and theoretical probability
- find the probability of complementary events



About the artwork

The pictures in this booklet are parts of the **Mandelbrot set**, drawn by a computer that repeats one simple rule, $z \rightarrow z^2 + c$, thousands of times for every point. Every booklet shows a different part of the set: this one is feathery spirals deep in Elephant Valley. Maths can make beautiful things.

Acknowledgement of Country

We acknowledge the Bundjalung people, the Traditional Custodians of the land on which Kingscliff High School stands, and pay our respects to Elders past and present.

NAME _____ CLASS _____ DATE _____

WE DO

Watch your teacher.
Copy every step.

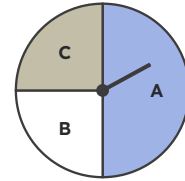
KEY IDEA

The sample space is the list of every possible outcome. Outcomes are equally likely when each one has the same chance of happening.

E1 List the sample space for rolling a die.

E2 List the sample space for tossing two coins.

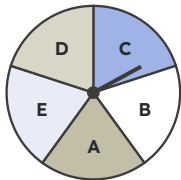
E3 Are the outcomes on this spinner equally likely?



Easy

Questions 1–6

1 List the sample space for this spinner.



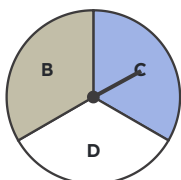
2 A letter is chosen from the word SCHOOL. List the sample space.

3 A counter is picked from a bag of blue, white, green counters. List the sample space.

4 A number from 1 to 12 is chosen. List the outcomes that are even.

5 How many outcomes are there when you roll a die?

6 List the sample space for this spinner.



Easy

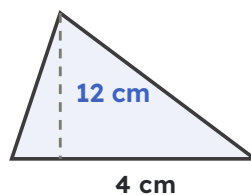
Questions 7–12

7 A letter is chosen from the word PENCIL. List the sample space.

8 A counter is picked from a bag of white, yellow, black counters. List the sample space.

9 A number from 1 to 8 is chosen. List the outcomes that are even.

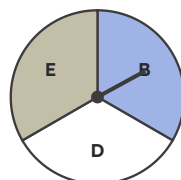
10 Find the area. (Not to scale.)



from 8.06

11 How many outcomes are there when you toss a coin?

12 List the sample space for this spinner.

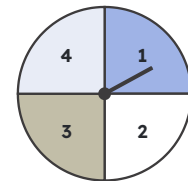


Medium

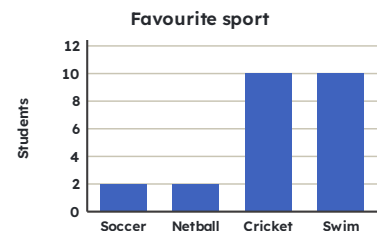
Questions 13–16

13 A bag has 3 green, 1 black and 8 yellow counters. Are the colours equally likely to be picked? Explain.

14 Are the outcomes on this spinner equally likely?



15 How many more students chose Swim than Soccer?



from 10.01

16 List the sample space for tossing a coin and rolling a die.

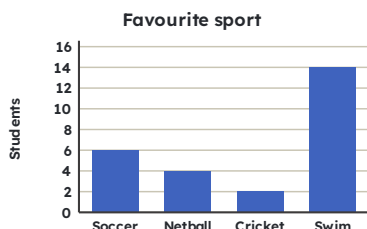
Medium

Questions 17–22

- 17 A bag has 7 red, 4 yellow and 6 black counters. Are the colours equally likely to be picked? Explain.

- 18 Two dice are rolled and the numbers are added. List the possible totals. Are they equally likely?

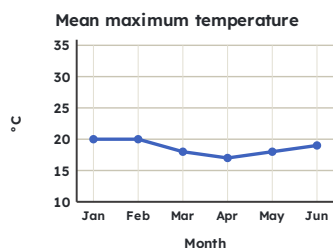
- 19 Which sport was the most popular?



from 10.01

- 20 A bag has 2 black, 1 blue and 3 yellow counters. Are the colours equally likely to be picked? Explain.

- 21 In which month was it warmest? By how much did it change from January to June?



from 10.01

- 22 A bag has 7 blue, 8 green and 5 white counters. Are the colours equally likely to be picked? Explain.

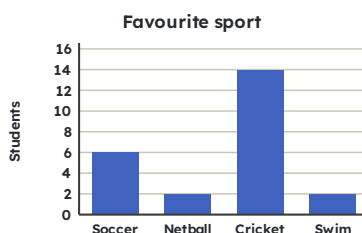
Challenging

Questions 23–25

- 23 List every order for the three letters B, A, C.

- 24 Two coins are tossed. How many outcomes are in the sample space? How many give one head and one tail?

- 25 How many students were surveyed altogether?



from 10.01

Challenging

Questions 26–25

NAME _____ CLASS _____ DATE _____

WE DO

KEY IDEA

When the outcomes are equally likely, $P(\text{event}) = \frac{\text{number of outcomes in the event}}{\text{the total number of outcomes}}$. Write it as a fraction, a decimal or a percentage.

Watch your teacher.
Copy every step.

E1 Find $P(\text{rolling an even number})$ on a die.

E2 A bag has 3 red and 5 blue counters. Find $P(\text{red})$.

E3 Find $P(\text{a heart})$ from a full deck of 52 cards.

Easy

Questions 1–9

1 A die is rolled. Find $P(3)$.

2 A bag has 2 white and 6 red counters. One is picked. Find $P(\text{white})$.

3 A die is rolled. Find $P(\text{an odd number})$.

4 A die is rolled. Find $P(1)$.

5 A bag has 2 blue and 4 yellow counters. One is picked. Find $P(\text{blue})$.

6 A die is rolled. Find $P(\text{a 1 or a 6})$.

7 A die is rolled. Find $P(6)$.

8 A bag has 8 yellow and 1 black counters. One is picked. Find $P(\text{yellow})$.

9 A letter is chosen from the word SCHOOL. List the sample space.

from 11.01

Easy

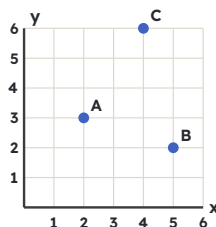
Questions 10–16

10 A bag has 5 yellow and 8 white counters. One is picked. Find $P(\text{yellow})$.

11 A die is rolled. Find $P(\text{a number greater than 4})$.

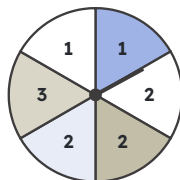
12 A die is rolled. Find $P(5)$.

13 Which point is at (5, 2)?



from 9.02

14 This spinner has equal sectors. Find $P(2)$.



___ out of ___
= $\frac{1}{3}$

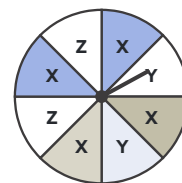
15 A bag has 6 white and 8 green counters. One is picked. Find $P(\text{white})$.

16 A die is rolled. Find $P(\text{a number greater than 2})$.

Medium

Questions 17–21

17 This spinner has equal sectors. Find $P(Y)$.



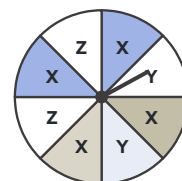
18 A card is picked from a full deck of 52. Find $P(\text{a black 7})$. Write it as a fraction and a percentage.

19 List the sample space for tossing a coin and rolling a die.

from 11.01

20 A letter is picked from APPLE. Find $P(P)$.

21 This spinner has equal sectors. Find $P(X)$.



Medium

Questions 22–28

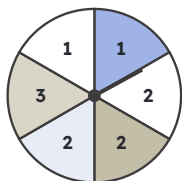
- 22 A card is picked from a full deck of 52. Find $P(\text{a heart})$. Write it as a fraction and a percentage.

- 23 A letter is picked from SUMMER. Find $P(E)$.

- 24 Write $\frac{9}{6}$ as a mixed numeral.

from 4.01

- 25 This spinner has equal sectors. Find $P(3)$.



- 26 A bag has 8 yellow, 7 green and 6 red counters. Are the colours equally likely to be picked? Explain.

from 11.01

- 27 Two coins are tossed. Find $P(\text{two heads})$ and $P(\text{one head and one tail})$.

- 28 A card is picked from a full deck of 52. Find $P(\text{the ace of spades})$. Write it as a fraction and a percentage.

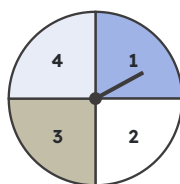
Challenging

Questions 29–37

- 29 Two dice are rolled and the numbers added. Find $P(\text{a total of } 12)$.

- 30 A bag has 2 yellow, 2 green and 3 white counters. Find $P(\text{yellow or green})$.

- 31 Are the outcomes on this spinner equally likely?



from 11.01

- 32 Two dice are rolled and the numbers added. Find $P(\text{a total of } 7)$.

- 33 Where does $y = x - 5$ cross the y -axis?

from 9.06

- 34 A bag has 5 yellow, 5 green and 7 red counters. Find $P(\text{yellow or green})$.

- 35 Two dice are rolled and the numbers added. Find $P(\text{a total of } 6)$.

- 36 Write $5\frac{11}{12}$ as an improper fraction.

from 4.01

- 37 A bag has 6 white, 7 red and 1 green counters. Find $P(\text{white or red})$.

Challenging

Questions 38–46

- 38 Two dice are rolled and the numbers added. Find $P(\text{a total of } 8)$.

- 39 A bag has 7 white, 5 blue and 6 green counters. Find $P(\text{white or blue})$.

- 40 A bag has 1 green, 1 black and 3 yellow counters. Are the colours equally likely to be picked? Explain.

from 11.01

- 41 A letter is picked from BANANA. Find $P(B)$.

- 42 Find the area of a triangle with base 4.4 m and height 14 m.

from 8.06

- 43 Two dice are rolled and the numbers added. Find $P(\text{a total of } 10)$.

- 44 A bag has 3 green, 5 yellow and 8 red counters. Find $P(\text{green or yellow})$.

- 45 Simplify $\frac{6}{8}$.

from 4.01

- 46 A letter is picked from APPLE. Find $P(L)$.

NAME _____ CLASS _____ DATE _____

WE DO

KEY IDEA

Probability goes from 0 (impossible) to 1 (certain). $\frac{1}{2}$ is an even chance. Below $\frac{1}{2}$ is unlikely; above $\frac{1}{2}$ is likely.

Watch your teacher.
Copy every step.

E1 Describe a probability of 0.8 in words.

E2 Describe the chance of rolling a 7 on a die.

E3 Order from least to most likely: rolling an even number, rolling a 6, rolling less than 7.

Easy

Questions 1–9

1 Describe a probability of 0.7 in words.

2 Describe the chance of rolling a 7 on a die.

3 What probability means impossible?

4 Describe a probability of 0.95 in words.

5 Describe the chance of the sun rising tomorrow.

6 What probability means an even chance?

7 Describe a probability of 0.1 in words.

8 Describe the chance of picking a red card from a deck.

9 A bag has 5 white and 2 green counters. One is picked. Find P(white).

from 11.02

Easy

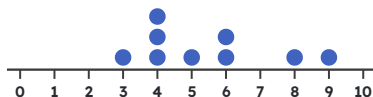
Questions 10–17

10 Describe the chance of rolling a 6 on a die.

11 What probability means certain?

12 Describe a probability of 0.05 in words.

13 Find the mode.



from 10.07

14 A probability is $\frac{7}{8}$. Is the event unlikely, an even chance or likely? Write it as a decimal.

$$\frac{7}{8} = \underline{\hspace{2cm}}$$

15 Describe the chance of a coin landing on heads.

16 Describe a probability of 1 in words.

17 A letter is chosen from the word BANANA. List the sample space.

from 11.01

Medium

Questions 18–22

18 A probability is $\frac{1}{6}$. Is the event unlikely, an even chance or likely? Write it as a decimal.

19 Order from least to most likely: rolling a 9 on a die; tossing a head; rolling a 6 on a die; rolling a number less than 5.

20 A letter is picked from PENCIL. Find P(P).

from 11.02

21 A probability is $\frac{5}{6}$. Is the event unlikely, an even chance or likely? Write it as a decimal.

22 Order from least to most likely: tossing a head; rolling a 6 on a die; rolling a number less than 5; rolling a 9 on a die.

Medium

Questions 23–25

- 23 A probability is $\frac{1}{10}$. Is the event unlikely, an even chance or likely? Write it as a decimal.

- 24 Describe an event that is unlikely but possible.

- 25 A counter is picked from a bag of white, red, black counters. List the sample space.

from 11.01

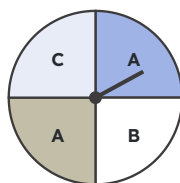
Challenging

Questions 26–31

- 26 Zac says an event has a probability of 1.2. Explain why that cannot be right.

- 27 Kai says an event has a probability of -0.2 . Explain why that cannot be right.

- 28 This spinner has equal sectors. Find $P(A)$.



from 11.02

- 29 Lena says an event has a probability of 2. Explain why that cannot be right.

- 30 A prism has a volume of 60 m^3 and a cross-section of 12 m^2 . How long is it?

from 8.12

- 31 Ali says an event has a probability of 1.5. Explain why that cannot be right.

Challenging

Questions 32–31

NAME _____ CLASS _____ DATE _____

WE DO

Watch your teacher.
Copy every step.

KEY IDEA

Experimental probability = the number of times the event happened ÷ the number of trials. With more trials it usually gets closer to the theoretical probability.

E1 A coin is tossed 50 times and lands heads 28 times. Find the experimental P(heads).

E2 What is the theoretical P(heads)? Compare.

E3 A die is rolled 120 times. How many 3s do you expect?

Easy

Questions 1–8

1 A coin is tossed 20 times and lands heads 8 times. Find the experimental P(heads).

2 A die is rolled 60 times. How many 6s would you expect?

3 Noah took 25 free throws and scored 14. Find the experimental probability of scoring, as a percentage.

4 A coin is tossed 20 times and lands heads 11 times. Find the experimental P(heads).

5 A die is rolled 300 times. How many 6s would you expect?

6 Tom took 20 free throws and scored 8. Find the experimental probability of scoring, as a percentage.

7 A coin is tossed 50 times and lands heads 21 times. Find the experimental P(heads).

8 A die is rolled 120 times. How many 6s would you expect?

Easy

Questions 9–16

9 A die is rolled 600 times. How many 6s would you expect?

10 Ali took 10 free throws and scored 9. Find the experimental probability of scoring, as a percentage.

11 A coin is tossed 20 times and lands heads 12 times. Find the experimental P(heads).

12 Where does $y = x + 6$ cross the y-axis?

from 9.06

13 A die is rolled 30 times and shows six 11 times. Compare the experimental and theoretical P(six).

experimental: $\frac{11}{30} \approx \frac{1}{3}$

theoretical: $\frac{1}{6} \approx \frac{1}{6}$

14 Aisha took 10 free throws and scored 4. Find the experimental probability of scoring, as a percentage.

15 A coin is tossed 20 times and lands heads 13 times. Find the experimental P(heads).

16 Write 9 out of 11 as a fraction in simplest form.

from 4.12

Medium

Questions 17–21

17 A die is rolled 90 times and shows six 18 times. Compare the experimental and theoretical P(six).

18 A spinner has $P(\text{blue}) = \frac{1}{4}$. How many blues do you expect in 40 spins?

19 Order from least to most likely: tossing a head; rolling a number less than 5; rolling a 6 on a die; rolling a 9 on a die.

from 11.03

20 A die is rolled 60 times and shows six 13 times. Compare the experimental and theoretical P(six).

21 A spinner has $P(\text{blue}) = \frac{1}{4}$. How many blues do you expect in 80 spins?

Medium

Questions 22–28

- 22 A spinner has $P(\text{blue}) = \frac{2}{3}$.
How many blues do you expect in 90 spins?

- 23 A die is rolled 30 times and shows six 4 times. Compare the experimental and theoretical $P(\text{six})$.

- 24 A letter is chosen from the word SCHOOL. List the sample space.

from 11.01

- 25 A die was rolled. Find the experimental $P(6)$. Do you think the die is fair? Explain.

Face	1	2	3	4	5	6
Count	8	10	10	10	10	26

- 26 A letter is picked from TEACHER. Find $P(A)$.

from 11.02

- 27 A spinner has $P(\text{blue}) = \frac{1}{4}$.
How many blues do you expect in 120 spins?

- 28 A die is rolled 60 times and shows six 6 times. Compare the experimental and theoretical $P(\text{six})$.

Challenging

Questions 29–34

- 29 A die was rolled. Find the experimental $P(6)$. Do you think the die is fair? Explain.

Face	1	2	3	4	5	6
Count	10	9	12	12	9	22

- 30 Jack tossed a coin 10 times and got 4 heads. Another student tossed it 100 times and got 47 heads. Whose result is a better estimate of $P(\text{heads})$? Why?

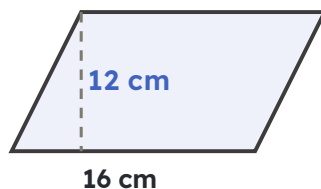
- 31 A probability is $\frac{2}{5}$. Is the event unlikely, an even chance or likely? Write it as a decimal.

from 11.03

- 32 A die was rolled. Find the experimental $P(6)$. Do you think the die is fair? Explain.

Face	1	2	3	4	5	6
Count	11	10	11	9	9	24

- 33 Find the area. (Not to scale.)



from 8.07

- 34 Chloe tossed a coin 10 times and got 4 heads. Another student tossed it 100 times and got 46 heads. Whose result is a better estimate of $P(\text{heads})$? Why?

Challenging

Questions 35–41

- 35 Aisha tossed a coin 10 times and got 7 heads. Another student tossed it 100 times and got 46 heads. Whose result is a better estimate of $P(\text{heads})$? Why?

- 36 A die was rolled. Find the experimental $P(6)$. Do you think the die is fair? Explain.

Face	1	2	3	4	5	6
Count	12	11	9	8	12	27

- 37 Change 2 m^3 to cm^3 .

from 8.10

- 38 A spinner has $P(\text{blue}) = \frac{3}{5}$.
How many blues do you expect in 150 spins?

- 39 Write 1 out of 8 as a fraction in simplest form.

from 4.12

- 40 Tom tossed a coin 10 times and got 4 heads. Another student tossed it 100 times and got 50 heads. Whose result is a better estimate of $P(\text{heads})$? Why?

- 41 A die was rolled. Find the experimental $P(6)$. Do you think the die is fair? Explain.

Face	1	2	3	4	5	6
Count	10	11	11	12	8	28

NAME _____ CLASS _____ DATE _____

WE DO

KEY IDEA

The complement of an event is the event not happening. An event and its complement add to 1, so $P(\text{not } A) = 1 - P(A)$.

Watch your teacher.
Copy every step.

E1 Find $P(\text{not rolling a 6})$ on a die.

E2 $P(\text{rain}) = 0.3$. Find $P(\text{no rain})$.

E3 What is the complement of rolling an even number?

Easy

Questions 1–9

1 $P(A) = 0.8$. Find $P(\text{not } A)$.

2 A die is rolled. Find $P(\text{not } 5)$.

3 What is the complement of rolling a 6?

4 $P(A) = 0.9$. Find $P(\text{not } A)$.

5 A die is rolled. Find $P(\text{not } 1)$.

6 What is the complement of picking a red card?

7 $P(A) = 0.5$. Find $P(\text{not } A)$.

8 A die is rolled. Find $P(\text{not } 2)$.

9 A bag has 6 white and 3 blue counters. One is picked. Find $P(\text{white})$.

from 11.02

Easy

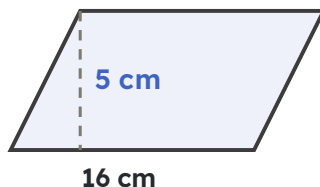
Questions 10–18

10 A die is rolled. Find $P(\text{not } 3)$.

11 What is the complement of winning a game (no draws)?

12 $P(A) = 0.4$. Find $P(\text{not } A)$.

13 Find the area. (Not to scale.)



from 8.07

14 The chance of rain is 28%. What is the chance of no rain?

___% - ___%

= ___%

15 A die is rolled. Find $P(\text{not } 6)$.

16 What is the complement of tossing a head?

17 Describe the chance of the sun rising tomorrow.

from 11.03

18 $P(A) = 0.2$. Find $P(\text{not } A)$.

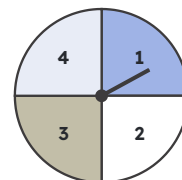
Medium

Questions 19–24

19 The chance of rain is 83%. What is the chance of no rain?

20 A bag has 6 white, 8 red and 4 yellow counters. Find $P(\text{not white})$.

21 Are the outcomes on this spinner equally likely?



from 11.01

22 $P(\text{win}) = \frac{5}{12}$. There are no draws. Find $P(\text{lose})$.

23 The chance of rain is 33%. What is the chance of no rain?

24 A spinner has $P(\text{blue}) = \frac{3}{5}$. How many blues do you expect in 150 spins?

from 11.04

Medium

Questions 25–33

- 25 A bag has 8 white, 8 yellow and 6 green counters. Find $P(\text{not white})$.

- 26 $P(\text{win}) = \frac{2}{7}$. There are no draws. Find $P(\text{lose})$.

- 27 In the table, what is y when $x = 2$?

x	1	2	3	4
y	5	8	11	14

from 9.04

- 28 The chance of rain is 45%. What is the chance of no rain?

- 29 A counter is picked from a bag of white, green, red counters. List the sample space.

from 11.01

- 30 A spinner lands on red, blue or green. $P(\text{red}) = 0.2$ and $P(\text{blue}) = 0.3$. Find $P(\text{green})$.

- 31 A bag has 7 green, 8 white and 5 yellow counters. Find $P(\text{not green})$.

- 32 A die is rolled. Find $P(\text{a number greater than 2})$.

from 11.02

- 33 $P(\text{win}) = \frac{4}{9}$. There are no draws. Find $P(\text{lose})$.

Challenging

Questions 34–41

- 34 A spinner lands on red, blue or green. $P(\text{red}) = 0.3$ and $P(\text{blue}) = 0.4$. Find $P(\text{green})$.

- 35 In a class of 50, 25 students walk to school. A student is picked. Find $P(\text{does not walk})$, as a percentage.

- 36 Find the area of a triangle with base 8.9 m and height 10 m.

from 8.06

- 37 Mia says " $P(\text{rolling a 2}) = \frac{1}{6}$ and $P(\text{rolling a 3}) = \frac{1}{6}$, so they are complementary." Is that right? Explain.

- 38 Aisha took 25 free throws and scored 17. Find the experimental probability of scoring, as a percentage.

from 11.04

- 39 A spinner lands on red, blue or green. $P(\text{red}) = 0.4$ and $P(\text{blue}) = 0.3$. Find $P(\text{green})$.

- 40 In a class of 40, 11 students walk to school. A student is picked. Find $P(\text{does not walk})$, as a percentage.

- 41 A bag has 3 red, 8 black and 8 blue counters. Are the colours equally likely to be picked? Explain.

from 11.01

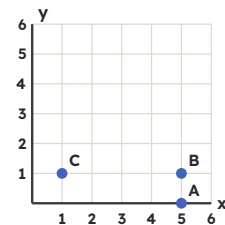
Challenging

Questions 42–49

- 42 In a class of 40, 3 students walk to school. A student is picked. Find $P(\text{does not walk})$, as a percentage.

- 43 A spinner lands on red, blue or green. $P(\text{red}) = 0.3$ and $P(\text{blue}) = 0.5$. Find $P(\text{green})$.

- 44 Which point is at (5, 0)?



from 9.02

- 45 $P(\text{win}) = \frac{3}{8}$. There are no draws. Find $P(\text{lose})$.

- 46 How many outcomes are there when you roll a die?

from 11.01

- 47 In a class of 30, 15 students walk to school. A student is picked. Find $P(\text{does not walk})$, as a percentage.

- 48 A spinner lands on red, blue or green. $P(\text{red}) = 0.3$ and $P(\text{blue}) = 0.3$. Find $P(\text{green})$.

- 49 Describe a probability of 0.5 in words.

from 11.03